

Basic Information

Experiment No.: 06

Date: 23-09-2026

Title: Java List Interface using ArrayList

Objective / Aim

To implement the List interface using "ArrayList" and demonstrate operations such as adding, accessing, updating, removing, searching, creating sublists, and sorting elements.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
-

Algorithm / Logic Description

1. Create a "List" using "ArrayList".
2. Add elements using "add()".
3. Insert an element at a specific position using "add(index, element)".
4. Access an element using "get()".

Source Code

```
package collectionframework;

import java.util.List;
import java.util.ArrayList;
import java.util.Comparator;

public class ListExample {
    public static void main(String[] args) {

        List<String> list = new ArrayList<>();

        // add()
        list.add("Apple");
        list.add("Banana");
        list.add("Mango");
        list.add("Apple");

        System.out.println("List: " + list);

        // add(index, element)
        list.add(1, "Orange");
        System.out.println("After adding at index 1: " + list);

        // get()
        System.out.println("Element at index 2: " + list.get(2));

        // set()
        list.set(2, "Grapes");
```

```
System.out.println("After set: " + list);

// remove(index)
list.remove(1);
System.out.println("After remove: " + list);

// indexOf()
System.out.println("First Apple index: "
    + list.indexOf("Apple"));

// lastIndexOf()
System.out.println("Last Apple index: "
    + list.lastIndexOf("Apple"));

// subList()
System.out.println("SubList: "
    + list.subList(0, 2));

// sort()
list.sort(Comparator.naturalOrder());
System.out.println("Sorted List: " + list);
}
}
```

Compilation & Execution Commands

Save the file as:

ListExample.java

Since the program contains:

```
package collectionframework;
```

the file should be inside the "collectionframework" folder.

Compile:

```
javac ListExample.java
```

Run:

```
java collectionframework.ListExample
```

Output Screen

List: [Apple, Banana, Mango, Apple]

After adding at index 1: [Apple, Orange, Banana, Mango, Apple]

Element at index 2: Banana

After set: [Apple, Orange, Grapes, Mango, Apple]

After remove: [Apple, Grapes, Mango, Apple]

First Apple index: 0

Last Apple index: 3

SubList: [Apple, Grapes]

Sorted List: [Apple, Apple, Grapes, Mango]

Result

The List interface was successfully implemented using "ArrayList", and operations such as "add()", "get()", "set()", "remove()", "indexOf()", "lastIndexOf()", "subList()", and "sort()" were successfully demonstrated.